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Abstract

The primary objective of this project is to analyse and compare various machine learning algorithms for predicting customer churn in the telecommunications industry. In the telecommunications industry, it is crucial to retain consumers in order to achieve sustainable corporate development due to the growing competition. The project aims to utilise sophisticated machine learning techniques to create prediction models that can effectively detect clients who are likely to churn. The dataset includes a range of customer characteristics, such as demographics, subscription services, and tenure, which is a valuable source of information for study. Through exploratory data analysis and feature engineering, the project tries to discover important factors that influence churn behaviour. Subsequently, a range of machine learning algorithms including decision trees, random forests, standard vector machine and XGBoost will be employed to build predictive models. The performance of these models will be analysed and compare based on metrics such as accuracy, precision, recall, and F1-score. The findings of this project are expected to contribute to the telecommunications industry by providing actionable insights for developing targeted retention strategies.

Chapter 1 : Introduction

1.1 Background

The telecommunications sector is a vital industry that facilitates global connectivity through various services such as mobile communication, internet access, and telephony. Telecommunication is essential for linking individuals, companies, and institutions worldwide, facilitating immediate communication and information sharing regardless of geographical boundaries. Telecommunication has become an essential component of daily life in our interconnected society, playing a crucial role in promoting economic growth, social development, and technical innovation. The widespread use of smartphones, fast internet, and digital communication platforms has completely transformed the way in which individuals interact, work, and access information.

Telecommunication infrastructure forms the backbone of modern economies, facilitating commerce, trade, and financial transactions on a global scale. Advanced communications networks allow organisations to reach consumers and partners on different continents, perform remote operations, and have access to a wide range of digital services and resources. Telecommunication also plays an important part in emergency response, public safety, and disaster management by providing critical communication channels during times of crisis or natural catastrophes. In this highly competitive market, customer satisfaction considers as the most important factor as it directly impacts customer loyalty, retention, and brand reputation. Also, customers expect customised goods and improved services at lower rates, and they are constantly searching and switching for better services from other providers (Lu, 2002).

Customer churn refers to the phenomenon where customers terminate their subscriptions or switch to competitors, resulting in revenue loss and decreased market competitiveness. In this dynamic and highly competitive sector, reducing customer churn holds paramount importance for sustaining business growth and profitability. Customer churn is a significant concern for companies in the telecommunications industry. One key reason why customer churn is crucial for telecommunications companies is its direct impact on revenue preservation. Losing customers means losing their subscription fees and potential future revenue from additional services or upgrades. Consequently, effective churn management is essential for maintaining a steady revenue stream and sustaining business operations.

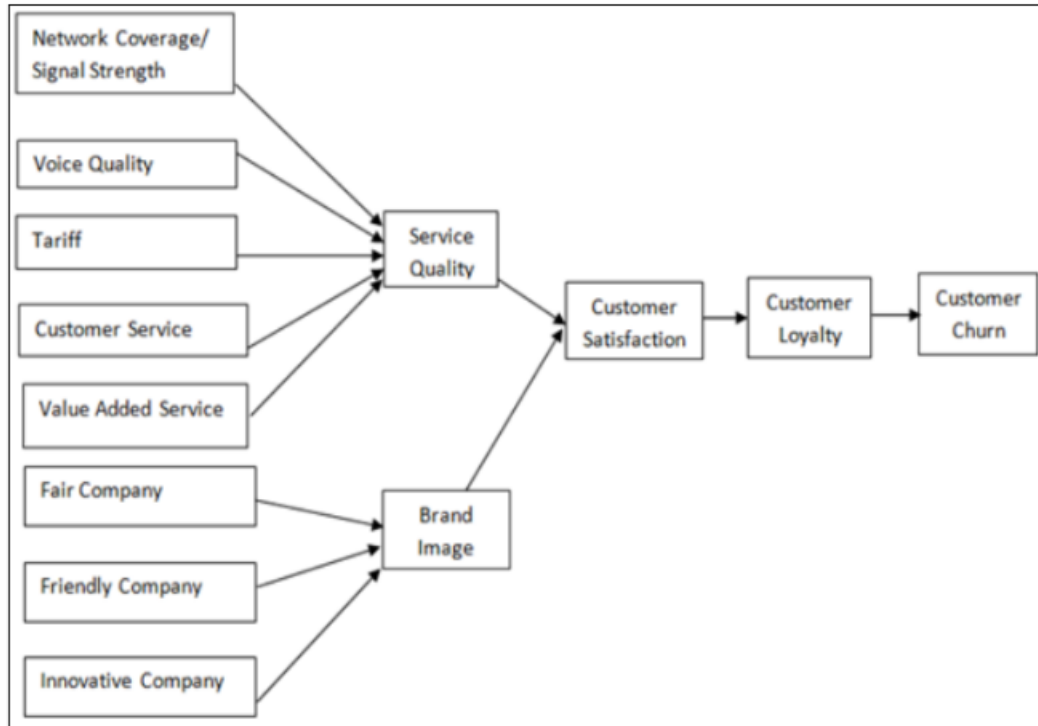


Figure 1: A framework on the background of customer churn (Misra, 2017).

Satisfied customers are more likely to remain loyal, renew subscriptions, and recommend services to others, thereby driving revenue growth and maintaining market competitiveness. Additionally, positive customer experiences contribute to enhanced brand reputation. Besides, reducing churn can lead to substantial cost savings for telecommunications companies. Acquiring new customers typically requires higher costs than retaining existing ones. By implementing strategies to mitigate churn, companies can minimize expenses related to marketing, sales, and customer acquisition efforts, thereby optimizing operational efficiency and resource allocation.

customer churn prediction plays a pivotal role in helping telecommunications companies proactively address churn risks. Predictive models enable companies to identify customers with a high likelihood of churning before they actually move from the service. By analysing churn patterns companies gain valuable insights into customer preferences, issues encountering, and satisfaction levels. This foresight empowers companies to implement targeted retention strategies.

1.2 Problem Statement

The telecommunications industry faces a persistent challenge in customer churn, where subscribers terminate their contracts or switch to competitors, resulting in revenue loss and diminished market share. Identifying customers who are likely to churn before they actually

do is crucial for implementing targeted retention strategies and minimizing the negative impact of churn on business performance. However, predicting customer churn accurately remains a complex task due to the multitude of factors influencing customer behaviour, including service quality, pricing, competition, and customer satisfaction.

Moreover, the telecommunications industry operates in a highly dynamic and competitive environment, characterized by rapid technological advancements and evolving customer preferences. As such, traditional approaches to churn prediction, such as rule-based systems or simple statistical models, may no longer suffice in capturing the intricacies of customer behaviour and predicting churn accurately. The objective of this study is to leverage machine learning algorithms and predictive analytics to accurately identify customers at risk of churn within the telecommunications sector. By analysing collected customer data, including usage patterns, demographic and service information, the goal is to build efficient churn prediction models that can assist telecommunication companies in proactively retaining customers and mitigating revenue losses.

1.3 Research Questions

The project is aims to answer the following questions:

- i. Which are the main features that contributes to the customer churn in the telecommunications sector?
- ii. Which is the efficient machine learning algorithm for the customer churn prediction in telecom industry?

1.4 Aim & Objectives

The overall goal of this project is to create an efficient customer churn prediction model designed exclusively for the telecoms sector. This will use robust machine learning model that accurately predicts instances of customer, allowing telecoms companies to identify and retain customers who are likely to leave their services. By capturing insights from previous customer data, the goal is to build a prediction tool that helps telecom companies to deploy targeted retention strategies.

The main objectives of this research are as follows:

- Gather relevant data pertaining to customer interactions, contractual dynamics, usage patterns and other demographic details from the telecommunications dataset.
- Identify and select relevant features that may influence customer churn.

- Conduct exploratory data analysis to gain insights into the relationships between various features and customer churn.
- Evaluate different machine learning algorithms suitable for churn prediction and select the most appropriate one based on performance metrics.
- Train and assess the performance of the trained model using cross-validation techniques to ensure its robustness and generalizability.
- Compare & analyse the models to find out the best and efficient machine learning model for customer churn prediction.

Chapter 2 : Literature Review

2.1 Customer Churn

Customer churn defined as the behaviour of a customer to leave or to terminate a service (Makhtar, 2017). Churn poses significant financial and strategic implications for companies operating within this sector. This literature review aims to examine the complex feature of customer churn in the telco industry, outlining its significance and exploring the methodologies employed for its prediction and mitigation.

According to (GoCardless, 2022) we can split customer churn into two Voluntary churn and Involuntary churn. Involuntary churn refers to the situation where a customer discontinues the use of your service without making an intentional choice to terminate their relationship with you. Voluntary churn refers to the situation where a consumer makes a conscious choice to terminate your service.

Customer churn serves as an indicator of customer dissatisfaction and market volatility within the telco domain. As customers opt to terminate their subscriptions or switch to rival providers, telco companies face not only immediate revenue losses but also enduring impacts on brand loyalty and market share. With the cost of acquiring new customers far be greater than that of retaining existing ones (Adnan Amin, 2017), the consequences of churn extend beyond mere financial metrics and impact market competitiveness and long-term growth.

2.2 Factors Influencing Customer Churn

Studies explore the various factors that influence customer churn in the telecom sector. These factors may include service quality, pricing, customer demographics, contract terms, customer satisfaction, and competition within the industry. Understanding these factors is crucial for developing effective churn prediction models.

According to the (Mustafa, 2021) analysis the customers with low Net Promoter Score (NPS) showing an elevated customer churn rate compared to other customers. Based on the findings, several key factors affecting low NPS, and churn include service quality, pricing, customer satisfaction, contract terms, and competition.

(Banda, 2017) conducted a comparative study on factors influencing mobile telecommunications customer churn in Zambia. (Banda, 2017) Cited among the 20 factors identified, poor customer service emerged as the primary cause, contributing to approximately 23% of the cases of mobile customer churn, with issues related to poor network quality and high service costs following closely.

(Khan, 2019) conducted a similar study about customer churn in Pakistan telecommunication services market and found service quality emerges as a critical factor, encompassing aspects such as network reliability, call clarity, and internet speed. Poor service quality often leads to customer dissatisfaction, prompting them to explore alternative service providers. (Khan, 2019) suggest that integrating value-added services into their offerings, telecommunication companies can not only meet but exceed customer expectations, developing greater loyalty and reducing the likelihood of customer attrition.

2.3 Importance of Customer Churn Analysis and Prediction

According to researchers, winning new customers in today's competitive conditions is up to 10 times costlier than retaining existing customers (Çelik, 2019). Churn analysis provides insights into customer behaviour and preferences, enabling companies to tailor their products and services to meet evolving customer needs. Ultimately, effective churn analysis not only helps reduce customer defection but also drives business growth and enhances competitiveness in the marketplace. By analysing churn patterns and identifying factors contributing to customer attrition, companies can implement targeted retention strategies to mitigate churn and retain valuable customer relationships.

The study by (Xevelonakis, 2005) underscores the significance of understanding and predicting customer churn patterns to inform strategic decision-making within the industry. Telco firms can use advanced analytics and predictive modelling approaches to acquire insights into the causes that cause customers to leave and design specific tactics to prevent churn. Moreover, proficient churn analysis allows organizations to optimize the resources allocation and marketing efforts, guaranteeing the highest possible impact in retaining valuable consumers.

2.4 Machine Learning Methods

(A., 2016) conduct comparison survey on various machine learning algorithms for customer churn prediction. Notable approaches include a hybrid SVM model with recursive feature elimination and Naive Bayes Tree for rule generation. Hybrid neural network models,

combining back-propagation artificial neural networks and self-organizing maps, demonstrated improved prediction accuracy. Boosting algorithms were employed to enhance churn prediction by separating customers into clusters based on risk, and SVM models were improved using random sampling to address data imbalance. (A., 2016) concludes by highlighting the need for constant development of accurate prediction models and suggests the combination of SVM with boosting algorithms as a potential avenue for future research in the domain of churn prediction.

(Ahmad, 2019) proposed a feature engineering phase, utilizing a big data platform and employing four tree-based machine learning algorithms such as Decision Tree, Random Forest, GBM, and XGBOOST also to enhance predictive accuracy AdaBoost were employed from previous research. after comparing the models, the (Ahmad, 2019) concluded that the XGBOOST algorithm achieved the highest accuracy value that outperforming all other models.

(Vafeiadis, 2015)Explored the machine learning algorithms including logistic regression, decision trees, random forests, support vector machines, and artificial neural networks. The study compares the performance of these algorithms based on metrics such as accuracy, precision, recall, and F1-score. After rigorous evaluation, the results indicate that random forests outperform other machine learning techniques in terms of predictive accuracy and best for customer churn prediction. Random forests leverage the power of collective learning, combining multiple decision trees to produce more accurate and reliable predictions, making them the preferred choice for churn prediction tasks in the telecommunications industry.

Chapter 3 : Exploratory Data Analysis

3.1 Overview

The primary objective of this project is to compare different machine learning algorithms and find the best machine learning method to predict the customer churn in the Telecommunication industry. This can be achieved through various processes like data collection, data preprocessing, exploratory data analysis, model selection, model evaluation, prediction, and finally the result analysis and comparison.

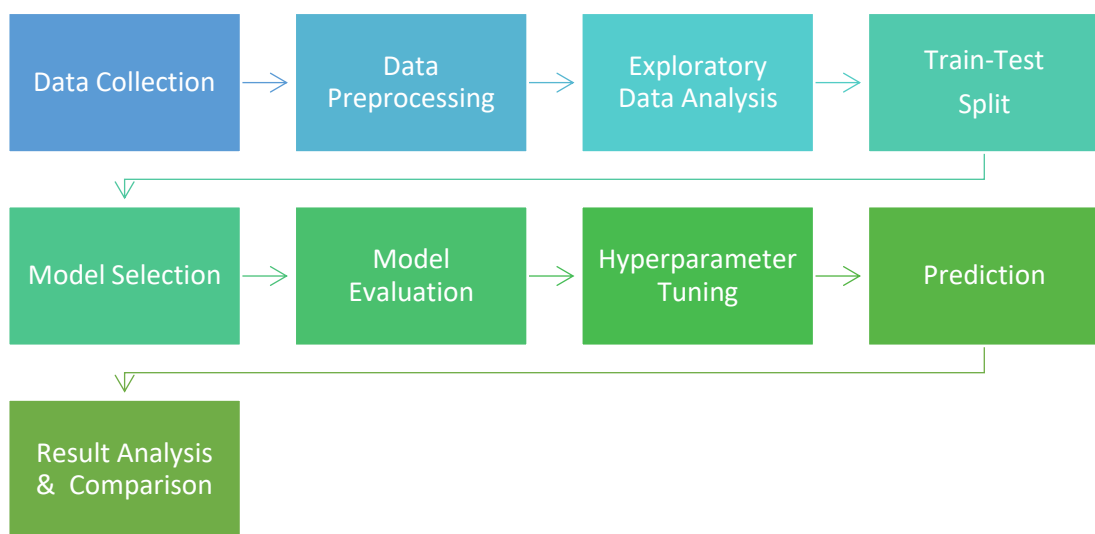


Figure 2: Project Overview

3.2 About the Dataset

The dataset used for this project is a comprehensive collection of Telco customer churn data that was obtained through Kaggle. It is commonly referred to as the IBM Watson dataset which consists of 7043 customers churn data. The dataset contains 21 attributes for each customer, including gender, tenure, phone service, online security, contract, and payment methods. It offers a comprehensive perspective on customer engagement by providing significant information on usage patterns, contractual behavior, and demographics. This dataset is a valuable resource for exploring the factors that influence customer churn.

Attribute	Data Type	Explanation
customerID	object	Unique id number for identifying each customer.
gender	object	The gender of the customer.
SeniorCitizen	int64	The customer is senior citizen or not.
Partner	object	Whether the customer has a partner or not.
Dependents	object	The customer has any dependent or not.
tenure	int64	How many months the customer subscribed to the services.
PhoneService	object	Whether customer has home phone service or not.
MultipleLines	object	Whether customer subscribed multiple line of phone services.
InternetService	object	What type of Internet service is subscribed by the customer - No, DSL, Fiber Optic, Cable.
OnlineSecurity	object	Whether customer subscribes to an additional online security service.
OnlineBackup	object	customer subscribes to any online backup service.
DeviceProtection	object	Customers have any device protection plan with the company.
TechSupport	object	Whether customer subscribed to any Tech support service.
StreamingTV	object	Customer subscribed to any TV Streaming services.

StreamingMovies	object	Customer subscribed to any Movie Streaming services.
Contract	object	customer's current contract type – Month to Month, One Year, Two Year.
PaperlessBilling	object	Customers choose paperless bill or not.
PaymentMethod	object	What type of payment method is customer used - Bank Withdrawal, Credit Card, Mailed Check.
MonthlyCharges	float64	The total monthly charge for the subscribed services.
TotalCharges	object	The customer's total charge for the subscribed period.
Churn	object	The customer churned or not that is whether they are continuing their subscription or not.

Table 1: Data attributes and its information

3.3 Dataset Ethical Considerations

The Telco Customer Churn dataset, obtained from Kaggle, exhibits ethical compliance in various aspects. The data collection process appears to have been conducted with proper permissions obtained from individuals, ensuring compliance with privacy regulations in addition, the dataset does not include any personally identifiable information, which helps address concerns about confidentiality and anonymity. Additionally, the data is anonymized, minimizing the risk of harm to individuals. It is reasonable to assume that the data has been collected ethically and with permission, as it is publicly available on a reputable platform like Kaggle. Moreover, the absence of personal information implies that individuals' rights and privacy are preserved, and ethical guidelines are upheld throughout the data usage process. Therefore, the ethical consequences of using the Telco Customer Churn dataset are minimal, confirming its appropriateness for study without causing any harm or ethical issues.

3.4 Data Preprocessing

Data preprocessing is an integral step in Machine Learning as the quality of data and the useful information that can be derived from it directly affects the ability of our model to learn (Kumar D. , 2021). Data preprocessing involves transforming raw data into a structured format that can be readily accessed by machine learning algorithms, hence enhancing the predictive performance.

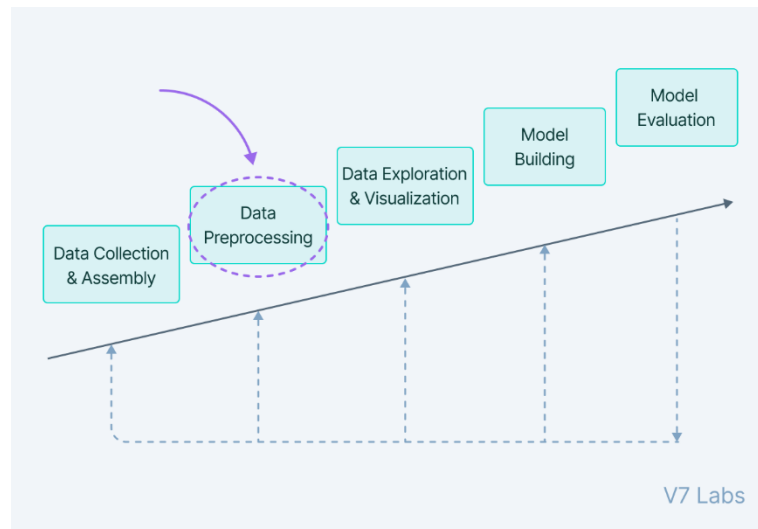


Figure 3: Data Preprocessing Step (Baheti, *A Simple Guide to Data Preprocessing in Machine Learning*, 2021)

3.4.1 Handling Missing value

Checking for the missing values across the dataset is one of the basic and mandatory preprocessing. Most of the datasets consist of NaN values or empty data in it. The cause of missing values can be data corruption or failure to record data (Kumar S. , 2021). Depending on the nature and amount of the missing data found on the dataset we can employ different strategies to handle the missing data. The common strategies used for handling the missing values are, dropping the rows with null values, filling missing values with constant values or mathematical measures like mean, median, mode, replace null values using series specific method etc. When the missing values are negligible compared to the total amount of data available in the dataset the common adoptable method is removing the rows with null values. In this project the missing values were associated with the tenure data which is numerical. So, the respective missing values are handled by filling it with the mean, median of the remaining available data.

3.4.2 Categorical Encoding

Categorical encoding is the process where all the categorical values are converted into the format that can be used by machine learning algorithms. Most of the machine learning algorithms numerical input for the prediction and analysis process. So, the categorical encoding process converts all the non-numerical data into numerical values. There are several types of encoding methods available for this conversion process like One-Hot encoding, Label encoding, Ordinal encoding etc. In this project most of the data is available in object datatype. The One-Hot encoder is used to transform these object data to numerical data and increase the performance of the machine learning algorithms.

3.4.3 Feature Selection

Feature Selection is the process where it removes irrelevant or redundant features from the dataset. This method helps to improve accuracy and reduce overfitting. The main purpose of

this approach is to select the feature the contribute to the predictive power of the model while removing the irrelevant features. Feature selection also helps speed up the training and prediction time.

3.4.4 Feature Scaling

Feature scaling is the process of setting the variables on a similar scale (Galli, 2023). Common scaling techniques include normalization, standardization or scaling to the minimum and maximum values. Standardization is a frequently employed technique for scaling features in machine learning applications. Standardization is the process of centering the variable at 0 and standardizing the variance to 1 (Galli, 2023). This method is less affected by outliers compared to other feature scaling methods.

$$\text{Standardized value} = \frac{(\text{Value} - \text{Mean})}{\text{Standard Deviation}}$$

3.5 Data Analysis

Exploratory Data Analysis is an important step towards understanding underlying patterns, correlations, and anomalies in a dataset. After the effective data preprocessing, the resulting data is immediately ready to use with machine learning algorithms. Gaining insights from the data is a crucial procedure that can provide guidance for feature engineering, preprocessing, and modeling.

3.5.1 Data Overview

After reading the data from the database you can get an overview of the dataset's structure and attributes by using two primary functions `shape` and `info()`.

The `shape` of the dataset indicates the number of rows and columns it contains. For example, the shape of (7043, 21) means there are 7043 samples and 21 features or columns in the dataset. This gives us an immediate understanding of the dataset's size and dimensionality.

```
Dataframe shape: (7043, 21)
Number of instances = 7043
Number of attributes = 21
```

Figure 4: Dataset Overview

The `info()` method offers a brief summary of the dataset, presenting the data types of each column, the count of non-null values, and the amount of memory used. It is a fast approach

for detecting missing values and understanding the data type. This will help with the execution of effective preprocessing of data on the next stage.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                7043 non-null   object
2   SeniorCitizen         7043 non-null   int64
3   Partner               7043 non-null   object
4   Dependents            7043 non-null   object
5   tenure                7043 non-null   int64
6   PhoneService          7043 non-null   object
7   MultipleLines         7043 non-null   object
8   InternetService       7043 non-null   object
9   OnlineSecurity        7043 non-null   object
10  OnlineBackup          7043 non-null   object
11  DeviceProtection      7043 non-null   object
12  TechSupport           7043 non-null   object
13  StreamingTV           7043 non-null   object
14  StreamingMovies       7043 non-null   object
15  Contract              7043 non-null   object
16  PaperlessBilling      7043 non-null   object
17  PaymentMethod         7043 non-null   object
18  MonthlyCharges        7043 non-null   float64
19  TotalCharges          7043 non-null   object
20  Churn                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

Figure 5: Dataset info

3.5.2 Understanding Data with describe()

The describe() generates a summary of the descriptive statistics for each numerical column in a Dataframe. describe() method, which provides a summary of the central tendencies, dispersion, and shape of a dataset's distribution (Cloud, 2023). For each numerical column it provides count, mean, standard deviation, minimum, 25%, 50% (median), 75%, and maximum information. Count indicates the number of non-null observation in that column, Mean gives the average value of the data points, minimum and maximum indicates the range in which the data falls, standard deviation indicates how the data spread around the mean. There are three quartile values—a lower quartile, median, and upper quartile—which divide the data set into four ranges, each containing 25% of the data points. It helps in understanding the data's spread and identifying potential outliers.

	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000
mean	32.371149	64.761692	2281.916928
std	24.559481	30.090047	2265.270398
min	0.000000	18.250000	18.800000
25%	9.000000	35.500000	402.225000
50%	29.000000	70.350000	1397.475000
75%	55.000000	89.850000	3786.600000
max	72.000000	118.750000	8684.800000

Figure 6: Data Description of Numerical Data

3.5.3 Insights from Graphs

In this section we will analyze various graphs and plots generated from the customer churn database. It will help to gain different useful insight that helps the business to understand their customer behavior that leads to the churning. Also, these insights help to take necessary strategies to retain their customers for a long period and improve the revenue.

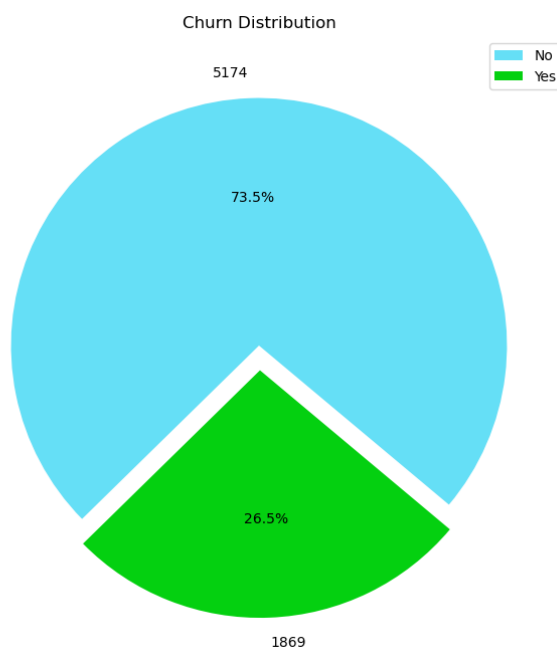


Figure 7: Churn Distribution

The above given pie chart shows the overall customer churn distribution based on the given data. It indicates that there is a noticeable proportion of customers who have left the service over the given time period. That is among the 7043 customers 73.5% (5174) of customers churned from the services due to several reasons. It means more than half of the customers are leaving the company within the timeframe, suggesting that the company is facing challenges to retain their customers. So, it is a very necessary situation to find an efficient customer retention strategy for the company's growth and profitability.

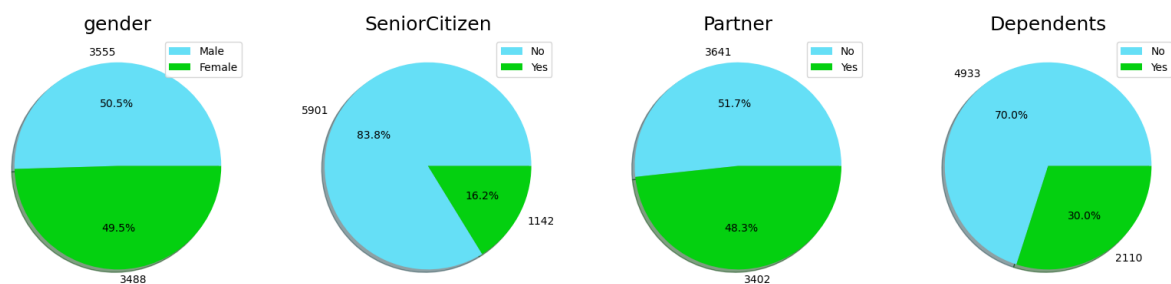


Figure 8: Demographic Feature Distribution

The above given graphs present the feature distribution and preferences of the Telco customers across various categorical variables. The distribution of gender is approximately equal, with a relatively equal number of male and female clients. In addition, a lesser percentage of consumers are senior citizens, suggesting that the service may be more popular among younger age groups. The majority of customers do not have a partner or dependents, indicating that the service is likely targeting more individual users rather than families or couples. This demographic insight reveals that the telecom service is predominantly used by single or individual customers.

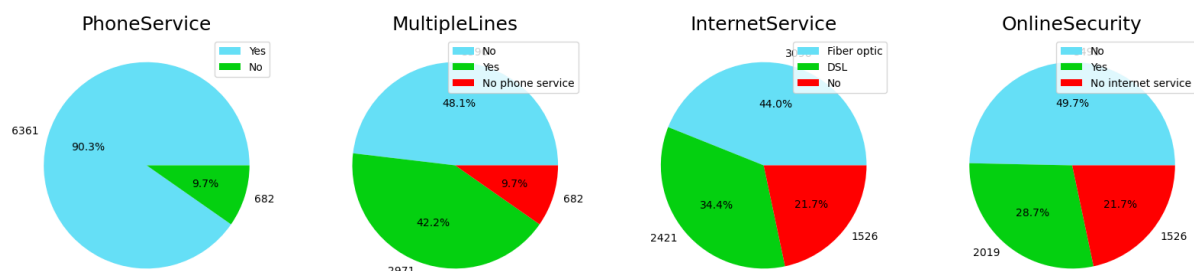


Figure 9: Service Feature Distribution

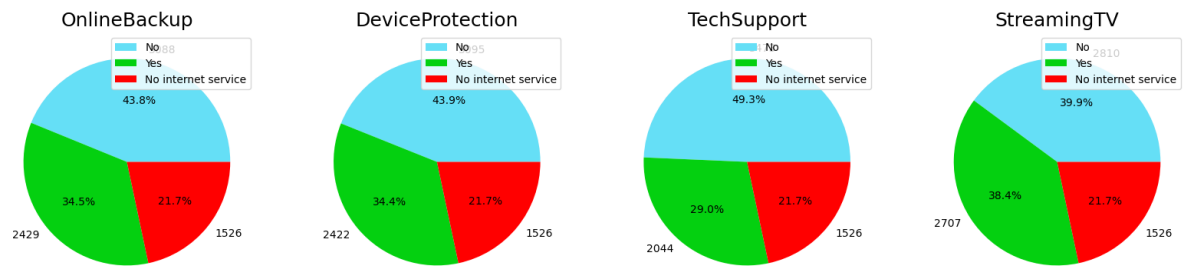


Figure 10: Security Feature Distribution

The dataset reveals that nearly all customers have phone service, which is to be expected considering the telecommunications nature of the service. Also, majority of the customers have multiple telephone lines and subscription to the internet. However, there are areas of concern. That is a large number of customers lack online security and backup services posing potential risks that could contribute to churn. Device protection and tech support is also notably absent among a significant segment of customers, presenting an opportunity for the company to introduce this as an additional service.

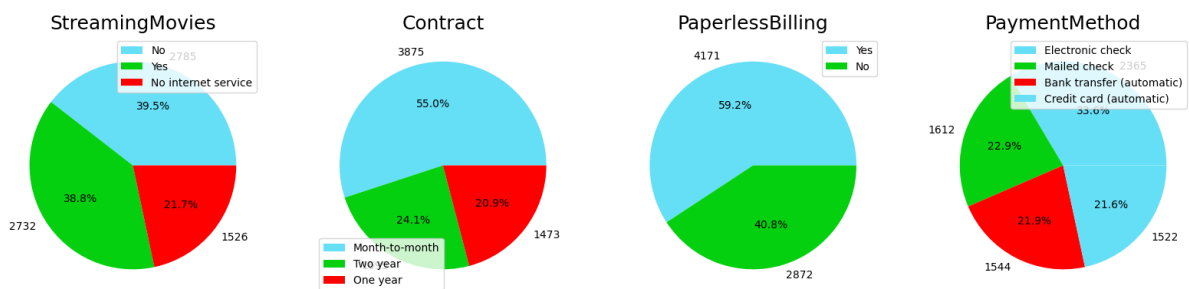


Figure 11: Payment Feature Distribution

A significant number of customers are actively using streaming TV and movie services, highlighting the importance of offering a wide range of entertainment content to maintain customer loyalty. When it comes to contract preferences, month-to-month contracts are the most popular choice among customers. This indicates that customers prioritize flexibility and are not interested in making long-term commitments. Also, the majority of customers opt for paperless billing, indicating a clear preference for digital and environmentally friendly billing methods. In terms of payment methods, electronic check emerges as the most favored option among the customers. This provides the organization with an opportunity to expand its payment options to better accommodate various customer preferences.

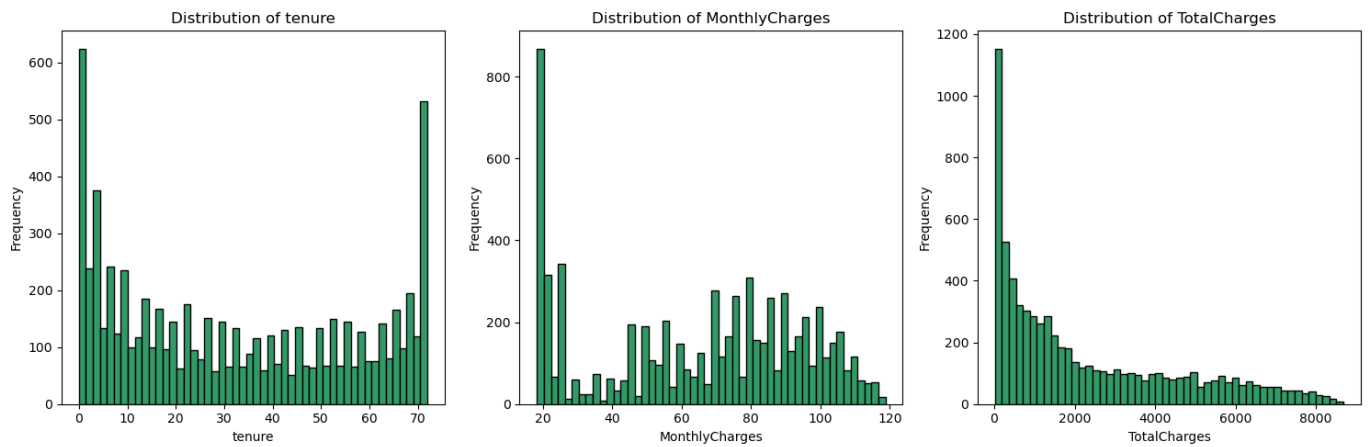


Figure 12: Numerical Data Distribution

The above given histogram provides insight about the distribution of numerical data that is Tenure, Monthly charges, Total charges. The distribution of tenure shows that a significant number of customers have been with the company for a relatively short period but the spread toward the higher tenure also indicates that company retains a portion of its customers over longer periods. Also, there is peak at the lower value may be indicating that large number of new customers. In the monthly charge distribution, the peak at lower monthly charges, indicating customers who opt for basic or discounted plans. The distribution of total charges appears to be right skewed with a majority of customers having lower total charges. This may be occurred because of the high number of new customers, or the customers prefer plans with low charge.

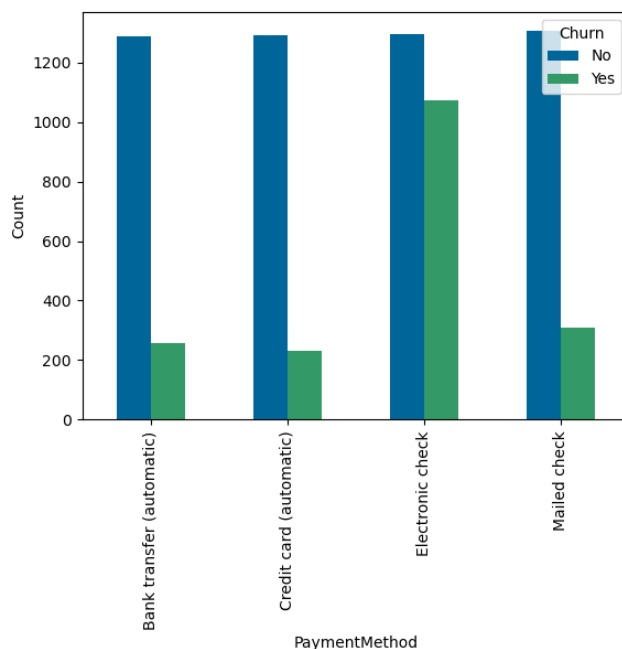


Figure 14: Payment Method Comparison

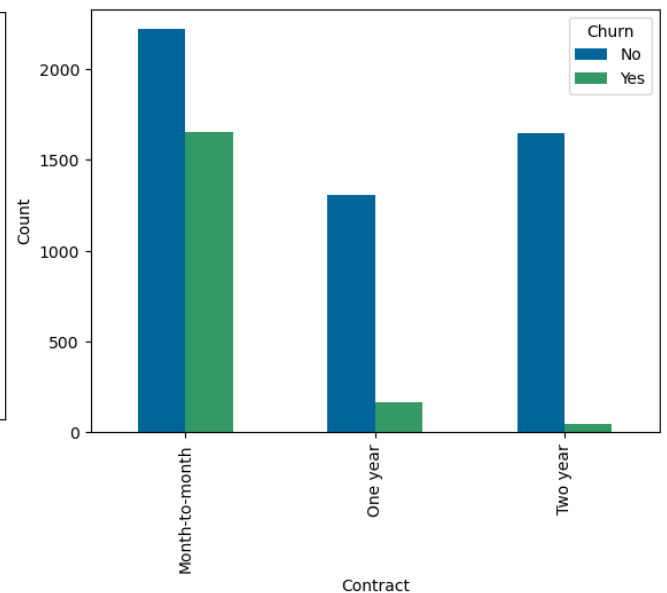


Figure 13: Contract Type Comparison

After analyzing all the attributes against the churn, we found out that Payment method and contract type are the two main features that contributes to the churn rate compared to all other

features. Customers using electronic check payment method show the highest churn rate compared to other methods. On the other hand, customers opting for bank transfer or credit card show relatively lower churn rates. This suggests that these payment methods might offer smoother transactions or additional benefits that encourage customer retention. The telecom company should consider improving the electronic check method or providing rewards to encourage the use of alternative payment methods. This could potentially decrease customer churn rates and increase customer satisfaction.

Customers who took month-to-month contracts exhibit a significantly high rate of churn in comparison to those who have one year or two year contracts. This suggests that customers on shorter-term contracts are more likely to churn, possibly due to less commitment or dissatisfaction with the service. Contrast customers with longer-term contracts show lower churn rates indicating that these longer contracts period might offer more stability. It implies that the telecommunications business could gain advantages by promoting long-term contracts or improving the offers of month to month plans in order to decrease customer churn.

Chapter 4 : Methodology

4.1 Project Environment

This project was developed in a Python environment, using various libraries and tools tailored for data analysis and machine learning tasks. The project environment was set up in Jupyter Notebook, which provided an interactive platform for coding and data exploration. We utilized the commonly used NumPy and Pandas libraries for data manipulation and preprocessing activities and Matplotlib.pyplot and Seaborn libraries are used for the visualization purpose that enabling us to create insightful charts and graphs to visualize the data's patterns and distributions. To address class imbalance in our dataset, we used the SMOTEEN technique from the imbalanced-learn library. Additionally, for implementing and evaluating machine learning models, we used the scikit-learn library along with XGBoost.

4.2 Classification Models

In this project, we used a comprehensive approach to address the problem of customer churn prediction. Initially we performed data preprocessing and exploratory data analysis to understand the characteristics of the dataset. In the next step we are going to find out the best customer churn predictive model by comparing the results of various machine learning algorithms. For our predictive modeling we implemented different machine learning algorithms including Decision Trees, Random Forest, Support Vector Machines (SVM), and XGBoost. After these algorithms were implemented, its performance were evaluated based on these four metrics, that is Accuracy, precision, recall and f1-score.

4.3 Training and Test

The train-test split is a technique used for evaluating the performance of a machine learning algorithm (Brownlee, 2020). Splitting dataset into train and test data is a fundamental step in develop and evaluating a machine learning model where train test used to train our models and test data confirms it works correctly with previously unseen data (Barkved, 2022). This approach helps in identifying potential issues like overfitting and provides a more realistic assessment of the model's performance on unseen data. A commonly used split ratio is 80:20, 70:30, 60:40, and 50:50 (Joseph, 2022). In this project we used an 80-20 split. It means 80% of the data was used for training the models, and 20% was kept for testing. This approach helps to ensure that the model is trained on a substantial amount of the data while leaving enough data for unbiased testing. Through this we can assess how well our models perform on new and unseen data. Before performing train test split, we need to randomize the data prevents any biases or patterns that might be influencing the model's performance. By shuffling the data we ensure that the training and testing sets are randomly sampled to provide more accurate evaluation of the model's performance.

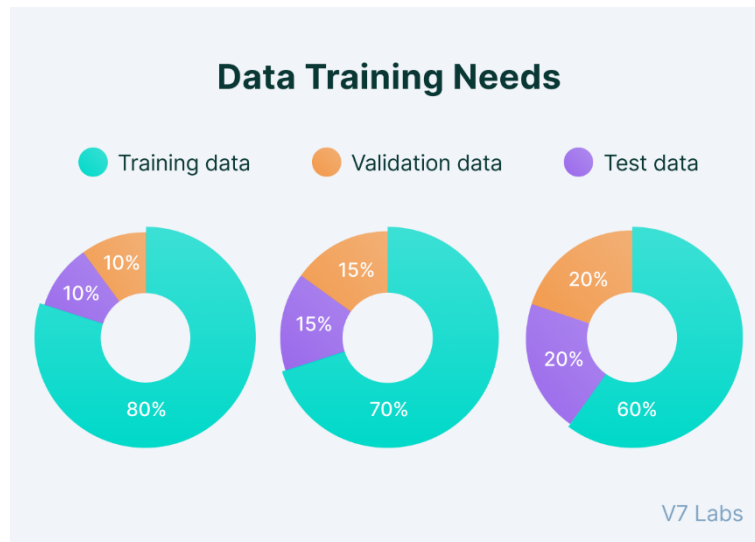


Figure 15: Training and test data (Baheti, *Train Test Validation Split: How To & Best Practices*, 2021)

4.4 Performance Metrics

4.4.1 Confusion Matrix

Confusion Matrix is a tabular visualization of the actual class labels versus model predictions (Bajaj, 2023). This matrix consists of true positive, true negative, false positive, false negative values. This matrix helps to identify where the model making mistakes like misclassifying the data.

- True Positive (TP): Number of correctly predicted positive classes.
- True Negative (TN): Number of correctly predicted negative classes.
- False Positive (FP): Number of incorrectly predicted positive classes.
- False Negative (FN): Number of incorrectly predicted negative classes.

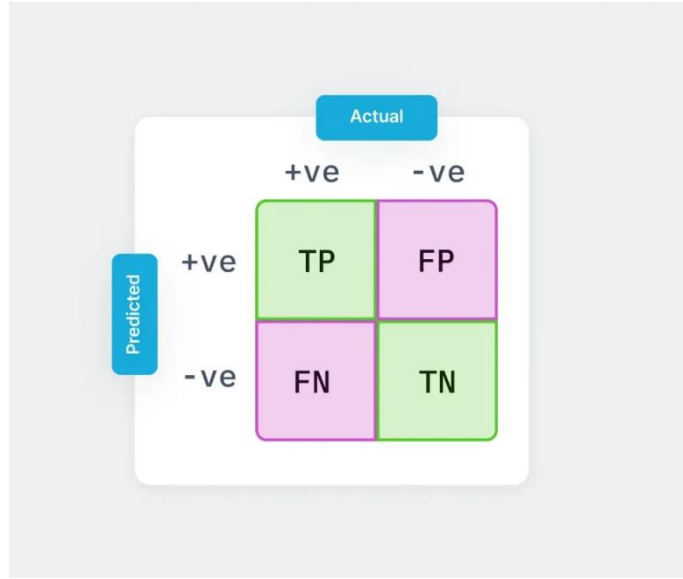


Figure 16: Confusion matrix (Shah, 2023)

4.4.2 Accuracy

Accuracy (A) is the primary performance metrics used to evaluate the overall performance of a classification model (Shah, 2023). It is the ratio of correctly predicted data to total number of data.

$$A = \frac{TP + TN}{TP + FP + TN + FN}$$

4.4.3 Precision

Precision (P) is a metric that measures the proportion of correctly predicted positive instances out of all the predicted positive instances (Jain, 2024). If the mode having high precision means it has lesser number of false positive values.

$$P = \frac{TP}{TP + FP}$$

4.4.4 Recall

Recall (R) also known as sensitivity or true positive rate (Shah, 2023). It is the ratio of positive instance to all actual positive in the dataset. Higher recall means lesser number of false negative.

$$R = \frac{TP}{TP + FN}$$

4.4.5 F1 Score

The F1 score combines recall and precision by calculating their weighted average. It is helpful while working with imbalanced dataset.

$$F1 = 2 * \frac{\text{Precision} * \text{recall}}{\text{Precision} + \text{recall}}$$

4.5 Hyperparameter Tuning

In machine learning models each models have a set of hyperparameters which are the external configuration variables that use to manage machine learning model training (What is Hyperparameter Tuning?, n.d.). Hyperparameter tuning will helps to optimize the models into its maximum strength. So, it is very crucial to choose the right hyper parameters for the models to give its maximum performance and accuracy. The optimization process consists of four main sections, an estimator with its objective function, a search space, an optimization method used to find hyper-parameter combinations, and an evaluation function to compare the performance of different hyper-parameter configurations (Shami, 2022). There are different types of hyper parameters that includes categorical, discrete, and continuous parameters and also, due to these different types of hyper-parameters the tuning process is different among the different methods (Yang, 2020).

Chapter 5 : Results and Comparison

5. 1 Methodology and Results

In this customer churn prediction project, we mainly focus on five different machine learning algorithms which includes, Decision tree, Random forest, Support vector machine, SVM with Adaboost and finally XGboost. In this section we are going to discuss the results we obtained through different ML methods and compare the result to find out the best method among these for the customer churn prediction in telcom industry.

5.1.1 Decision Tree

The Decision Tree model exhibited noticeable improvement post hyper-parameter tuning. Initially, the model had an accuracy of approximately 72.6%. Also, from the classification report we can obtain the information such as precision, recall, and F1-scores. It indicates that a slightly better performance in predicting non-churn instances.

Accuracy: 0.7260468417317246					
Classification Report:					
	precision	recall	f1-score	support	
0.0	0.81	0.81	0.81	1036	
1.0	0.48	0.49	0.49	373	
accuracy			0.73	1409	
macro avg	0.65	0.65	0.65	1409	
weighted avg	0.73	0.73	0.73	1409	

Figure 17: Decision Tree Performance Metrics

After the parameter optimization the accuracy of the model significantly improved to 80.6%. while the precision and f1- score for both non-churn and churn classes improved respectively the recall score for class1 increase to 90%, suggesting a better identification of non-churn instances, while for churn slightly decreased to 54%. Overall, the parameter tuning notably enhanced the Decision Tree model's ability to distinguish between churn and non-churn instances.

```
# optimized decision tree classifier
decision_tree_model = DecisionTreeClassifier(criterion = "gini", random_state = 42, max_depth=6, min_samples_leaf=8)
churn_predict(decision_tree_model, x_train, y_train, x_test, y_test)
```

Accuracy: 0.8062455642299503

Classification Report:

	precision	recall	f1-score	support
0.0	0.84	0.90	0.87	1036
1.0	0.67	0.54	0.60	373
accuracy			0.81	1409
macro avg	0.76	0.72	0.73	1409
weighted avg	0.80	0.81	0.80	1409

Figure 18: Decision Tree Hyperparameter Tuning

5.1.2 Random Forest

After parameter tuning showed slight improvement in its accuracy. Initially, the model had an accuracy of about 79.55%. From the classification report we can see that similar to decision tree method in the case of random forest also showing stronger performance in predicting non-churn instances compared to churn.

Accuracy: 0.7955997161107168

Classification Report:

	precision	recall	f1-score	support
0.0	0.83	0.91	0.87	1036
1.0	0.66	0.47	0.55	373
accuracy			0.80	1409
macro avg	0.74	0.69	0.71	1409
weighted avg	0.78	0.80	0.78	1409

Figure 19: Random Forest Performance Metrics

For the hyper parameter tuning for the random forest method, we utilize the grid search technique that systematically searches through a specified grid of hyperparameters to find the optimal combination. The parameters identified through grid search were then applied to our random forest model, resulting in the maximum possible performance that is 81.19%. The tuned model revealed better precision and recall for predicting churn instances, suggesting that the tuning helped in capturing churn patterns more effectively.

```
# Define the Random Forest classifier with the best parameters obtained from grid search
best_rf_classifier = RandomForestClassifier(criterion='gini', n_estimators=100, max_depth=10, min_samples_split=2,
                                           min_samples_leaf=4, bootstrap=True)
churn_predict(best_rf_classifier,x_train, y_train, x_test, y_test)
```

Accuracy: 0.8119233498935415

Classification Report:

	precision	recall	f1-score	support
0.0	0.84	0.91	0.88	1036
1.0	0.69	0.53	0.60	373
accuracy			0.81	1409
macro avg	0.77	0.72	0.74	1409
weighted avg	0.80	0.81	0.80	1409

Figure 20: Random Forest Hyperparameter Tuning

5.1.3 Support Vector Machine

At first, l SVM model achieved an accuracy of 81.12% with the default hyperparameters. The precision for class churn was 0.69, recall was 0.52, and the F1-score was 0.59. It is also showing stronger performance in predicting non-churn class compared to churn.

Accuracy: 0.8112136266855926

Classification Report:

	precision	recall	f1-score	support
0.0	0.84	0.92	0.88	1036
1.0	0.69	0.52	0.59	373
accuracy			0.81	1409
macro avg	0.77	0.72	0.74	1409
weighted avg	0.80	0.81	0.80	1409

Figure 21: SVM Performance Metrics

For optimizing the SVM model, we utilized Grid Search method and applied the parameters obtained through the grid search. But for better optimization we changed the parameter kernel to linear which improved the accuracy from 81.4% to 82.1%.

```
best_svm_classifier =SVC(C=1, coef0=0.1, degree=2, gamma='scale', kernel='linear')
churn_predict(best_svm_classifier,x_train, y_train, x_test, y_test)
```

Accuracy: 0.8211497515968772

Classification Report:

	precision	recall	f1-score	support
0.0	0.86	0.90	0.88	1036
1.0	0.69	0.60	0.64	373
accuracy			0.82	1409
macro avg	0.77	0.75	0.76	1409
weighted avg	0.81	0.82	0.82	1409

Figure 22: SVM Hyperparameter Tuning

5.1.4 SVM with AdaBoost

According to various journals SVM with AdaBoost show an improved accuracy. To analyse this scenario, we implemented SVM with AdaBoost in the customer churn dataset. But at the end of the analysis AdaBoost provided a lesser accuracy of 76.5% compared to the support vector method.

```
Accuracy: 0.7650816181689141
Classification Report:
              precision    recall  f1-score   support

    0.0         0.89      0.77      0.83     1036
    1.0         0.54      0.74      0.63      373

 accuracy          0.77     1409
 macro avg         0.72     1409
 weighted avg      0.80     1409
```

Figure 23: SVM with AdaBoost

5.1.5 XGBoost

After analysing the XGBoost method we can see that initially before parameter tuning the model shows an accuracy of 78.934 %. also, precision for predicting churn was 64%, with a recall of 51% and F1-score of 57%. This indicates that while the model is reasonably good at identifying churn cases.

```
Accuracy: 0.7934705464868701
Classification Report:
              precision    recall  f1-score   support

    0.0         0.84      0.89      0.86     1036
    1.0         0.64      0.51      0.57      373

 accuracy          0.79     1409
 macro avg         0.74     1409
 weighted avg      0.78     1409
```

Figure 24: XGBoost performance metrics

For finding the best hyperparameter set for the XGBoost we used the search method. After applying parameter tuning with the parameters obtained through grid search to the XGBoost model, we observed the same accuracy of 79.34% with a slight improvement in other factors that is precision 64%, recall 51%. Consequently, F1-score 57%. Although the overall accuracy remained the same, the tuned model demonstrated a slightly better balance between precision and recall.

```
best_xgb_classifier = xgb.XGBClassifier(gamma= 0, learning_rate= 0.1, max_depth= 4, min_child_weight= 3, n_estimators= 50)
churn_predict(xgb_classifier,x_train, y_train, x_test, y_test)
```

Accuracy: 0.7934705464868701

Classification Report:

	precision	recall	f1-score	support
0.0	0.84	0.89	0.86	1036
1.0	0.64	0.51	0.57	373
accuracy			0.79	1409
macro avg	0.74	0.70	0.72	1409
weighted avg	0.78	0.79	0.79	1409

Figure 25: XGBoost Parameter Tuning

5.2 SMOTE

SMOTE, which stands for Synthetic Minority Over-sampling Technique, is a method used to handle class imbalance in datasets. SMOTE is specifically designed to tackle imbalanced datasets by generating synthetic samples for the minority class (SATPATHY, 2023). Class imbalance occurs when one class in the dataset has significantly more samples than the other class, which can lead to biased models that perform poorly on the minority class. Mainly there are two types of imbalances in dataset. One is between class imbalance, in which case some classes have much more samples than others and the second one is within-class imbalance, in which some subsets of one class have much few examples than other subsets of the same class.

```
Churn
No      73.463013
Yes     26.536987
Name: churn, dtype: float64
```

Figure 26: Percentage of Churn Distribution

In this project, SMOTE was used to address the class imbalance in the dataset. The dataset had a pointed imbalance between the number of churned customers and non-churned customers. This imbalance can lead to models that are biased towards predicting the majority class as they prioritise maximising overall accuracy while neglecting the performance of the minority class. Below tables show the values of performance metrics for each method before and after applying SMOTE.

	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.81	0.67	0.54	0.60
Random Forest	0.81	0.69	0.53	0.60
SVM	0.82	0.69	0.60	0.64

AdaBoost	0.77	0.54	0.74	0.63
XGBoost	0.79	0.64	0.51	0.57

Table 2: Model Performance Before Applying SMOTE

	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.93	0.91	0.96	0.94
Random Forest	0.96	0.95	0.97	0.96
SVM	0.92	0.91	0.94	0.92
AdaBoost	0.91	0.99	0.93	0.91
XGBoost	0.96	0.95	0.97	0.96

Table 3: Model Performance After Applying SMOTE

5.3 Result Comparison

After comparing the performance of all the five machine learning models that used in this project before applying SMOTE, we can see that SVM become the top performer with an accuracy of 82.11%. Random Forest came in second with an accuracy of 81.19%, followed by XGBoost and Decision Tree with 79.35% and 80.62%, respectively. Compared to all other machine learning methods AdaBoost performed the least with an accuracy 76.51%.

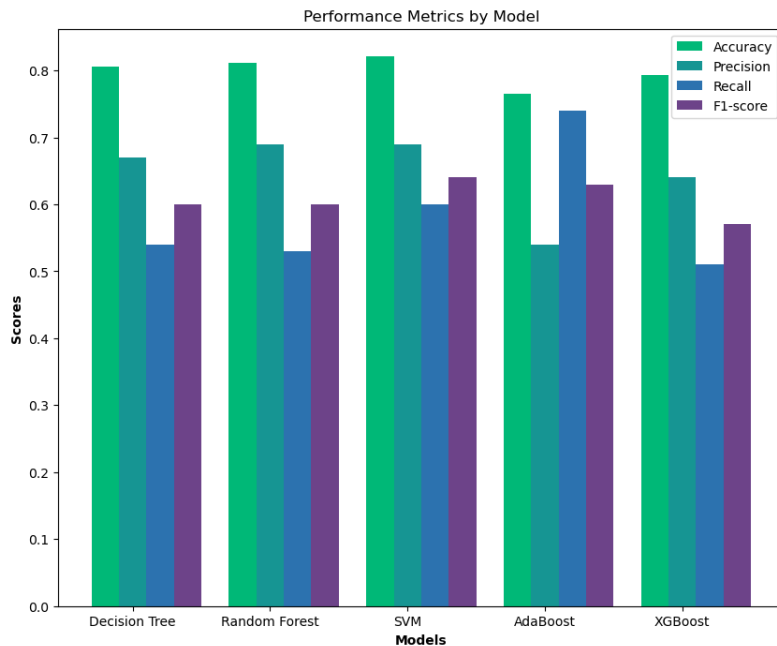


Figure 27: Performance metrics before SMOTE

While all models showed relatively competitive accuracies, SVM with highest accuracy proved to be the best method to predict customer churn in telecom industry. SVM may have performed better because of its versatility with different kernel functions and ability to find the optimal decision boundary between different classes. (Fagbuyiro, 2023).

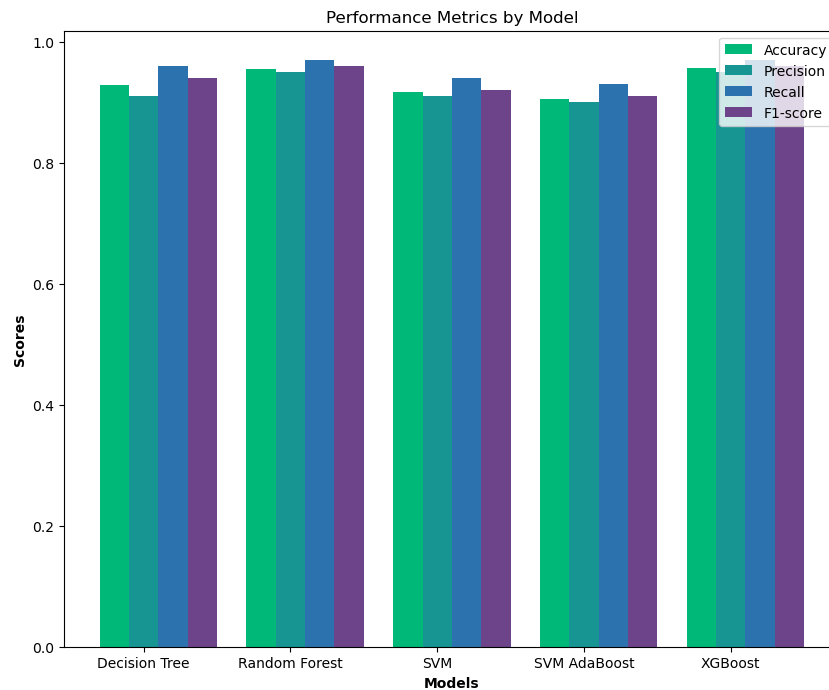


Figure 28: Performance Metrics After SMOTE

After addressing the data imbalance using SMOTE we observed significant improvements in the performance metrics across all models. Comparing all the model performance XGBoost had the highest accuracy of 95.65% Random Forest again came in second with an accuracy of 95.48%, followed by decision tree, SVM and AdaBoost with 92.8% and 91.7%, 90.6% respectively. In conclusion, after SMOTE implementation, XGBoost stands out as the most effective model for customer churn prediction in this dataset. This not only provide high accuracy but also exhibit a good balance between the features making them well suited for real world applications in customer churn prediction.

Chapter 6 : Conclusion

6.1 Summery

In this project we aimed to find out the best machine learning algorithm for customer churn prediction. So, we performed various machine learning algorithms and data analysis techniques to predict customer churn in a telecom industry. We started by exploring the dataset and understanding the key features that might influence churn. After that we applied different algorithms, including decision trees, random forests, SVM, AdaBoost, XGBoost, and compared its performance to find the most effective approach. By using hyperparameter tuning and techniques like SMOTE for handling class imbalance, we achieved improved results. Now it's the time for us to review our objectives to verify that we achieved all of them.

Objective 1: the main features that contributes to the customer churn in the telecommunications sector.

After conducting a detailed data exploration and analysis we found out that all the features have some effect on the customer churn. But among all the contract type and payment method plays a crucial role. So, by concentrating more on these two features will help the business to reduce the customer churn and help to retain their existing customers.

Objective 2: The efficient machine learning algorithm for the customer churn prediction in telecom industry.

We applied set of machine learning algorithms on balanced and imbalanced dataset. After the careful analysis based on various performance metrics we found out that SVM performed well in predicting customer churn on the imbalanced dataset with highest accuracy of 0.82. But after balancing the data with SMOTE we found out that XGBoost performed better than all other algorithms with an incredible accuracy of 0.96.

6.2 Ideas for future work

Future work on the telco customer churn prediction should be conducted on a more robust dataset which Integrate additional data sources such as customer support interactions, social media activity, or demographic information to improve the dataset and provide more framework for churn prediction. It will also help the business to analyse various complex situations to predict possible customer churn and create a customer retention plan. Experiment with more advanced machine learning algorithms or combined methods to further improved prediction accuracy. Models like Gradient Boosting Machines (GBM), Neural Networks, or even deep learning architectures could be explored. Further enhance the hyperparameters of the current models by employing advanced optimisation approaches such

as Bayesian optimisation or advanced algorithms to identify the most effective combination of parameters that maximise the model performance.

By exploring these future works, the customer churn prediction project can continue to expand and adapt to better solutions that suit the business needs, eventually leading to more accurate and practical insights to reduce customer churn.

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